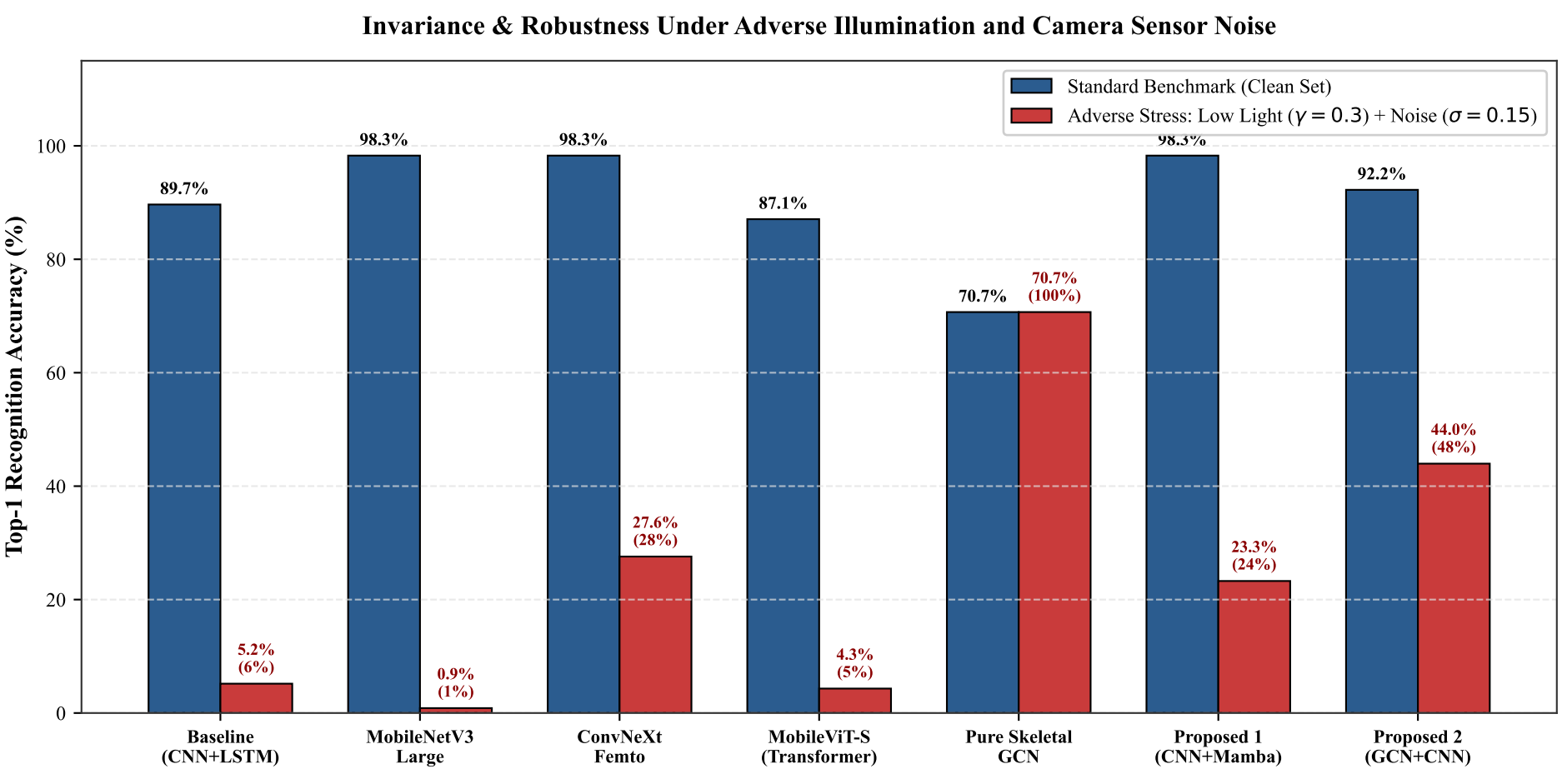
Invariance & Robustness Under Adverse Illumination and Camera Sensor Noise
Standard Benchmark (Clean Set)
Adverse Stress: Low Light ($\gamma = 0.3$) + Noise ($\sigma = 0.15$)
Top-1 Recognition Accuracy (%)
0
20
40
60
80
100
89.7%
5.2%
(6%)
98.3%
0.9%
(1%)
98.3%
27.6%
(28%)
87.1%
4.3%
(5%)
70.7%
70.7%
(100%)
98.3%
23.3%
(24%)
92.2%
44.0%
(48%)
Baseline
(CNN+LSTM)
MobileNetV3
Large
ConvNeXt
Femto
MobileViT-S
(Transformer)
Pure Skeletal
GCN
Proposed 1
(CNN+Mamba)
Proposed 2
(GCN+CNN)